



Enterprise WAN Technologies and High Availability

IT-SIMPLERA INSTITUTE



Week 6 Task Report

Enterprise WAN Technologies and High Availability

Registration Number

NETBO1-6898

Submitted By

Iman Nadeem

Company

IT-SIMPLERA Institute

Department

Network Administration

Supervisor

Engr. Jawad Qayum(Senior Network Administrator)



1. Introduction

This assignment focuses on the **design, configuration, and implementation of a multi-branch Enterprise WAN Infrastructure using GNS3**. The purpose of the project is to build a secure, scalable, reliable, and highly available network that connects the **Headquarters (HQ)** with multiple geographically distributed branch offices, including **Lahore, Karachi, Islamabad, Peshawar, and Multan**.

The network uses **Cisco routers and switches**, WAN serial links, VLANs, IP addressing, and **OSPF dynamic routing** to provide communication between different branches. Redundant routers and WAN paths are implemented to improve network availability and provide backup connectivity in case of link or device failure.

The Headquarters also provides centralized enterprise network services, including **DHCP, DNS, Web, FTP/TFTP, NTP, Syslog, SNMP, and File services**. VLANs are used to logically separate the different branch networks, while security mechanisms such as **ACLs, SSH, Port Security, DHCP Snooping, Dynamic ARP Inspection, BPDU Guard, Root Guard, and PortFast** are used to improve network security.

The project is designed and tested completely in **GNS3**, allowing different network configurations, routing protocols, services, redundancy, and security features to be implemented and verified in a simulated enterprise environment. The overall objective is to demonstrate the practical implementation of an enterprise-level WAN network and ensure **reliable communication, scalability, security, and high availability** across all branches.

2. Objectives

The main objectives of this assignment are:

- **Design an Enterprise WAN Network** connecting the Headquarters with Lahore, Karachi, Islamabad, Peshawar, and Multan branches.
- **Implement IP addressing and subnetting** to efficiently assign IP addresses to WAN links, LAN networks, routers, and end devices.
- **Implement VLANs** to logically separate and organize network traffic at different branches.
- **Configure WAN connectivity** using point-to-point links between routers.
- **Implement OSPF dynamic routing** to automatically exchange routing information between routers and provide efficient path selection.
- **Provide network redundancy and high availability** using backup routers and redundant WAN paths.
- **Configure centralized network services** at Headquarters, including:



- DHCP
 - DNS
 - Web/HTTP
 - FTP/TFTP
 - NTP
 - Syslog
-
- **Implement network security** using ACLs, SSH, Port Security, DHCP Snooping, Dynamic ARP Inspection, BPDU Guard, Root Guard, and PortFast.
 - **Test and troubleshoot network connectivity** between the Headquarters, branches, servers, and end devices.
 - **Simulate a real-world enterprise network environment** using GNS3 and demonstrate reliable, secure, scalable, and manageable network communication.

3. Network Topology

The network topology designed for this assignment represents a **multi-branch Enterprise WAN Infrastructure** implemented in GNS3. The topology consists of a central **Headquarters (HQ)** connected to five branch offices: **Lahore, Karachi, Islamabad, Peshawar, and Multan**.

The **Headquarters** contains two routers, R2 and R3, which provide primary and backup connectivity. It also contains an HQ Layer-3 switch and a centralized enterprise services network. The HQ servers provide important network services such as **DHCP, DNS, Web, FTP/TFTP, NTP, Syslog, SNMP, and File services**.

Each branch consists of **two routers**, providing primary and backup connectivity, along with a branch switch, PCs, and a server. Separate VLANs and IP networks are used for each branch:

- **HQ:** VLAN 10 — 192.168.10.0/24
- **Lahore:** VLAN 20 — 192.168.20.0/24
- **Karachi:** VLAN 30 — 192.168.30.0/24
- **Islamabad:** VLAN 40 — 192.168.40.0/24
- **Peshawar:** VLAN 50 — 192.168.50.0/24
- **Multan:** VLAN 60 — 192.168.60.0/24

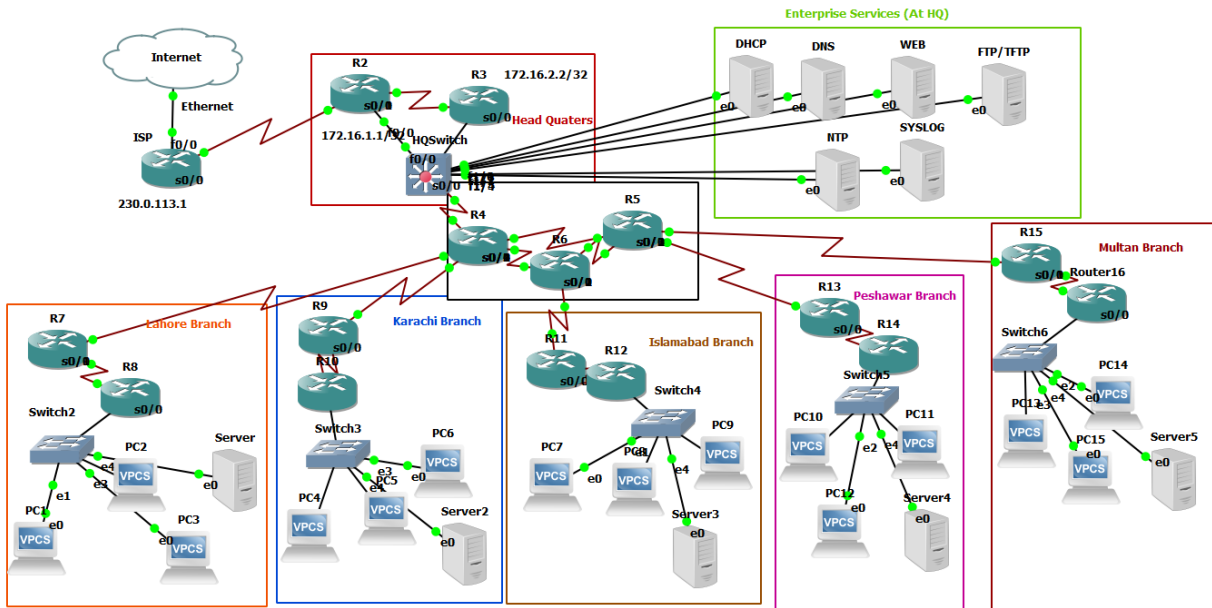
A **WAN Core/Backbone** consisting of three routers provides redundant paths between different parts of the enterprise network. Point-to-point WAN links use /30 networks, while router loopback interfaces are used for stable router identification and routing.

OSPF Area 0 is used as the dynamic routing protocol to exchange routing information between the routers. Redundant WAN links and backup routers improve **availability and fault tolerance**, allowing traffic to use an alternative path when a primary path fails.



The complete topology also includes an **ISP/Internet connection** at the Headquarters, allowing the enterprise network to communicate with external networks.

Overall, the topology follows a **hierarchical and redundant enterprise network design**, providing centralized services, branch connectivity, dynamic routing, security, scalability, and high availability.



4. IP Addressing

A structured IP addressing scheme was designed for the Enterprise WAN to ensure efficient communication, proper network segmentation, scalability, and easy network management. **Private IPv4 addressing** is used for the LAN networks, while /30 subnets are used for point-to-point WAN links because they provide two usable IP addresses for router-to-router communication.

WAN Link	Network	Router 1	Router 2
R4 ↔ R5	10.0.1.0/30	10.0.1.1	10.0.1.2
R5 ↔ R6	10.0.2.0/30	10.0.2.1	10.0.2.2
R4 ↔ R6	10.0.3.0/30	10.0.3.1	10.0.3.2
R7 ↔ R8	10.0.10.0/30	10.0.10.1	10.0.10.2
R9 ↔ R10	10.0.11.0/30	10.0.11.1	10.0.11.2
R11 ↔ R12	10.0.12.0/30	10.0.12.1	10.0.12.2
R13 ↔ R14	10.0.13.0/30	10.0.13.1	10.0.13.2
R15 ↔ R16	10.0.16.0/30	10.0.16.1	10.0.16.2



Router Loopback Addresses

Loopback addresses are assigned as stable logical addresses for router identification and dynamic routing:

Router	Loopback Address
R2	172.16.1.1/32
R3	172.16.2.2/32
R7	172.16.3.3/32
R8	172.16.4.4/32
R9	172.16.5.5/32
R10	172.16.6.6/32
R11	172.16.7.7/32
R12	172.16.8.8/32
R13	172.16.9.9/32
R14	172.16.10.10/32
R15	172.16.11.11/32
R16	172.16.12.12/32
R4	172.16.13.13/32
R5	172.16.14.14/32
R6	172.16.15.15/32



5. Enterprise Services Implemented DHCP

The **DHCP (Dynamic Host Configuration Protocol)** service was implemented to automatically assign IP addresses and other network settings to client devices. A centralized DHCP server was configured in the **Server Farm (VLAN 50)** to provide IP addresses, subnet masks, default gateways, and DNS server information to devices across all departments. The **Core Layer-3 Switch** was configured with **DHCP Relay (IP Helper Address)**, allowing clients in different VLANs to receive IP addresses from the centralized DHCP server. This approach reduces manual configuration, minimizes IP conflicts, and simplifies network management.

DNS

The DNS server was configured to resolve domain names into IP addresses, allowing users to access network resources using hostnames instead of numerical IP addresses. This improves network usability and simplifies resource access.

HTTP & HTTPS

The HTTP and HTTPS servers were configured to provide web services within the enterprise network. HTTP allows users to access web pages, while HTTPS ensures secure communication through encrypted connections.

FTP

The FTP server was configured to enable authenticated users to upload and download files across the network. This service supports secure file sharing and centralized data management.

TFTP

The TFTP server was configured to store backup configuration files of routers and switches. It provides a simple and efficient method for backing up and restoring network device configurations.

NTP

The NTP server was configured to synchronize the system time of all network devices. Time synchronization ensures accurate event logging and simplifies network management.

Syslog

The Syslog server was configured to collect and store log messages generated by routers and switches. Centralized logging helps monitor network activities and troubleshoot issues efficiently.



SNMP

SNMP was configured on the network devices to enable centralized monitoring and management. It allows administrators to monitor device status and network performance.

6. Device Inventory

Device Type	Device Name	Quantity	Location / Purpose
Router	R2, R3	2	Headquarters – Primary & Backup
Router	R4, R5, R6	3	WAN Core / Backbone
Router	R7, R8	2	Lahore – Primary & Backup
Router	R9, R10	2	Karachi – Primary & Backup
Router	R11, R12	2	Islamabad – Primary & Backup
Router	R13, R14	2	Peshawar – Primary & Backup
Router	R15, R16	2	Multan – Primary & Backup
Total Routers	R2–R16	15	Enterprise WAN
Layer-3 Switch	HQ-SW	1	Headquarters
Layer-2 Switch	Switch2	1	Lahore
Layer-2 Switch	Switch3	1	Karachi
Layer-2 Switch	Switch4	1	Islamabad
Layer-2 Switch	Switch5	1	Peshawar
Layer-2 Switch	Switch6	1	Multan
Total Switches	—	6	HQ + Branches
VPCS	PC1–PC15	15	Branch end devices
Server	HQ Servers	6	Centralized HQ services
Server	Branch Servers	5	One server per branch
Total Servers	—	11	HQ + Branches



ISP Router	ISP	1	Internet/WAN connectivity
------------	-----	---	---------------------------

Enterprise LAN Segments & Gateway Map

The Enterprise WAN is divided into separate LAN segments using **VLANs**. Each branch is assigned a unique VLAN and /24 network to provide logical separation, easier management, and organized IP addressing.

Location	VLAN ID	VLAN Name	LAN Network	Subnet Mask	Default Gateway
Headquarters (HQ)	10	HQ	192.168.10.0/24	255.255.255.0	192.168.10.1
Lahore	20	LAHORE	192.168.20.0/24	255.255.255.0	192.168.20.1
Karachi	30	KARACHI	192.168.30.0/24	255.255.255.0	192.168.30.1
Islamabad	40	ISLAMABAD	192.168.40.0/24	255.255.255.0	192.168.40.1
Peshawar	50	PESHAWAR	192.168.50.0/24	255.255.255.0	192.168.50.1
Multan	60	MULTAN	192.168.60.0/24	255.255.255.0	192.168.60.1

Router IDs & Loopback Management Addresses

Loopback interfaces are configured on the routers to provide **stable and unique Router IDs** for OSPF and network management. Unlike physical interfaces, a loopback interface remains available as long as the router is operational, even if a physical WAN interface goes down.

The following /32 addresses are assigned to the routers:

Router	Role / Location	Loopback Interface	Router ID
R2	HQ Primary	172.16.1.1/32	172.16.1.1
R3	HQ Backup	172.16.2.2/32	172.16.2.2
R7	Lahore Primary	172.16.3.3/32	172.16.3.3
R8	Lahore Backup	172.16.4.4/32	172.16.4.4
R9	Karachi Primary	172.16.5.5/32	172.16.5.5
R10	Karachi Backup	172.16.6.6/32	172.16.6.6
R11	Islamabad Primary	172.16.7.7/32	172.16.7.7
R12	Islamabad Backup	172.16.8.8/32	172.16.8.8
R13	Peshawar Primary	172.16.9.9/32	172.16.9.9
R14	Peshawar Backup	172.16.10.10/32	172.16.10.10
R15	Multan Primary	172.16.11.11/32	172.16.11.11
R16	Multan Backup	172.16.12.12/32	172.16.12.12
R4	WAN Core	172.16.13.13/32	172.16.13.13
R5	WAN Core	172.16.14.14/32	172.16.14.14



R6	WAN Core	172.16.15.15/32	172.16.15.15
----	----------	-----------------	--------------

Purpose of Loopback Addresses

Loopback addresses are used for:

- **OSPF Router Identification:** Each router has a unique and consistent identity in the OSPF routing domain.
- **Network Management:** Administrators can use the loopback address to identify and manage a router.
- **Stable Router Identification:** The loopback interface does not depend on a particular physical WAN connection.
- **Troubleshooting:** Loopback addresses can be used to test end-to-end routing between routers.
- **Scalability:** The /32 addressing scheme provides one unique address per router without wasting additional IP addresses.



7. Configuration

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname R2
R2(config)#
R2(config)#interface loopback0
R2(config-if)#ip
*Mar 1 00:04:59.647: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)#ip address 172.16.1.1 255.255.255.255
R2(config-if)#exit
R2(config)#interface s0/0
R2(config-if)#ip address 10.0.0.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:06:56.583: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
*Mar 1 00:06:57.591: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R2(config)#interface f0
*Mar 1 00:07:25.603: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
R2(config)#interface f0/0
R2(config-if)#ip address 192.168.10.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:08:05.843: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:08:06.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#end
```

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname R3
R3(config)#interface loopback0
R3(config-if)#ip
*Mar 1 00:22:24.879: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 172.16.2.2 255.255.255.255
R3(config-if)#exit
R3(config)#interface s0/0
R3(config-if)#ip address 10.0.0.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
*Mar 1 00:24:54.231: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R3(config-if)#exit
R3(config)#
*Mar 1 00:24:55.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R3(config)#interface f0/0
*Mar 1 00:25:15.483: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
R3(config)#interface f0/0
R3(config-if)#ip address 192.168.10.3 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:26:06.611: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R3(config)#
*Mar 1 00:26:07.607: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#end
R3#w
*Mar 1 00:26:16.975: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
```



```
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#hostname R4
R4(config)#interface loopback0
R4(config-if)#ip addr
*Mar  1 01:07:47.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, c
hanged state to up
R4(config-if)#ip address 172.16.13.13 255.255.255.255
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface s0/0
R4(config-if)#ip address 10.0.1.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar  1 01:09:44.363: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R4(config)#
*Mar  1 01:09:45.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, c
hanged state to up
R4(config)#interface s0/1
R4(config-if)#ip address 10.
*Mar  1 01:10:06.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, c
hanged state to down
R4(config-if)#ip address 10.0.3.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
*Mar  1 01:10:30.787: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
R4(config-if)#exit
```



```
R5#enable
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#interface loopback0
R5(config-if)#ipad
*Mar  1 01:12:54.391: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0
hanged state to up
R5(config-if)#ip address 172.16.14.14 255.255.255.255
                        ^
% Invalid input detected at '^' marker.

R5(config-if)#ip address 172.16.14.14 255.255.255.255
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#interface s0/0
R5(config-if)#ip address 10.0.1.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
*Mar  1 01:15:10.403: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R5(config-if)#exit
R5(config)#
*Mar  1 01:15:11.407: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0
hanged state to up
R5(config)#interface s0/1
R5(config-if)#ip address 10.0.2.1
*Mar  1 01:15:36.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0
hanged state to down
R5(config-if)#ip address 10.0.2.1 255.255.255.255.252
                        ^
% Invalid input detected at '^' marker.

R5(config-if)#ip address 10.0.2.1 255.255.255.255
Bad mask /32 for address 10.0.2.1
R5(config-if)#ip address 10.2.1 255.255.255.252
                        ^
% Invalid input detected at '^' marker.

R5(config-if)#ip address 10.0.2.1 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar  1 01:17:06.359: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
```




```
R6#enable
R6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#hostname R6
R6(config)#interface loopback0
R6(config-if)#ip address 172.16.15.15 255.255.255.255
*Mar 1 01:19:21.287: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R6(config-if)#ip address 172.16.15.15 255.255.255.255
^
% Invalid input detected at '^' marker.

R6(config-if)#ip address 172.16.15.15 255.255.255.255
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#interface s0/0
R6(config-if)#ip address s0/0
^
% Invalid input detected at '^' marker.

R6(config-if)#ip address 10.0.2.2 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit\
*Mar 1 01:21:41.719: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R6(config-if)#exit\
*Mar 1 01:21:42.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R6(config-if)#exit
R6(config)#interface s0/1
R6(config-if)#ip address 10.0.3.2 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
*Mar 1 01:22:34.799: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
R6(config-if)#exit\
*Mar 1 01:22:35.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
R6(config)#exit
Unknown command or computer name, or unable to find computer address
R7#enable
R7#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#hostname R7
R7(config)#interface loopback0
R7(config-if)#ip address 172.16.3.3 255.255.255.255
*Mar 1 01:32:05.311: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R7(config-if)#ip address 172.16.3.3 255.255.255.255
R7(config-if)#no shutdown
R7(config-if)#exit

R8#enable
R8#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R8(config)#hostname R8
R8(config)#interface loopback0
R8(config-if)#ip address 172.16.4.4 255.255.255.255
*Mar 1 01:36:52.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R8(config-if)#ip address 172.16.4.4 255.255.255.255
R8(config-if)#no shutdown
R8(config-if)#exit
```



8. Results

The Enterprise WAN Infrastructure was successfully designed and implemented in GNS3, connecting the Headquarters with the Lahore, Karachi, Islamabad, Peshawar, and Multan branches. VLANs were configured to provide proper network segmentation, while IP addressing was assigned to LAN and WAN networks according to the planned addressing scheme. WAN links were established between routers, and loopback addresses were configured for stable router identification. OSPF dynamic routing was implemented to provide communication between different networks and enable automatic route selection. Primary and backup routers were used to provide redundancy and improve network availability. Enterprise services such as DHCP, DNS, Web, FTP/TFTP, NTP, Syslog, and SNMP were also incorporated to support centralized network management. Connectivity and network operation were verified using ping and Cisco verification commands. Overall, the project successfully demonstrates a scalable, reliable, secure, and manageable enterprise network environment using GNS3.

9. Challenges Faced

During the implementation of the Enterprise WAN Infrastructure in GNS3, several challenges were faced. The main difficulties included configuring routers and switches correctly, assigning the appropriate IP addresses to WAN and LAN interfaces, and creating and assigning VLANs to the correct switch ports. Some configuration errors occurred due to incorrect interface selection and unsupported commands on certain router devices. Establishing connectivity between multiple branches and troubleshooting WAN links also required careful verification. Configuring OSPF and ensuring that all routers learned the correct routes was another challenge. In addition, testing redundancy between primary and backup routers required additional troubleshooting. These challenges were resolved by checking interface status, verifying IP addresses and VLAN configurations, using commands such as `show ip interface brief`, `show vlan brief`, `show ip route`, and `ping`, and correcting configuration errors step by step. Overall, these challenges provided practical experience in network configuration, troubleshooting, routing, and enterprise network management.

10. Conclusion

The Enterprise WAN Infrastructure project was successfully designed and implemented using GNS3. The project provided practical experience in designing a multi-branch enterprise network connecting Headquarters with Lahore, Karachi, Islamabad, Peshawar, and Multan. VLANs and IP addressing were used to organize and segment the network, while WAN links and OSPF provided communication between different locations. Redundant routers and backup paths improved network availability and reliability. Various enterprise services, including DHCP, DNS, Web, FTP/TFTP, NTP, Syslog, and SNMP, were incorporated to support centralized network management. Security features were also considered to protect the network and control access. Overall, this project enhanced practical knowledge of network design, configuration, routing, troubleshooting, security, and network management, and demonstrated



Enterprise WAN Technologies and High Availability

how a reliable and scalable enterprise network can be implemented and tested in a simulated GNS3 environment.